

Investigating Robust Overfitting in Adversarial Training

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Adversarial Example: Panda to Gibbon (Goodfellow et al., 3)

[Enter Text Here]

[Enter Text Here]



x

“panda”

57.7% confidence

+ .007 ×



$\text{sign}(\nabla_x J(\theta, x, y))$

“nematode”

8.2% confidence

=



$x + \epsilon \text{sign}(\nabla_x J(\theta, x, y))$

“gibbon”

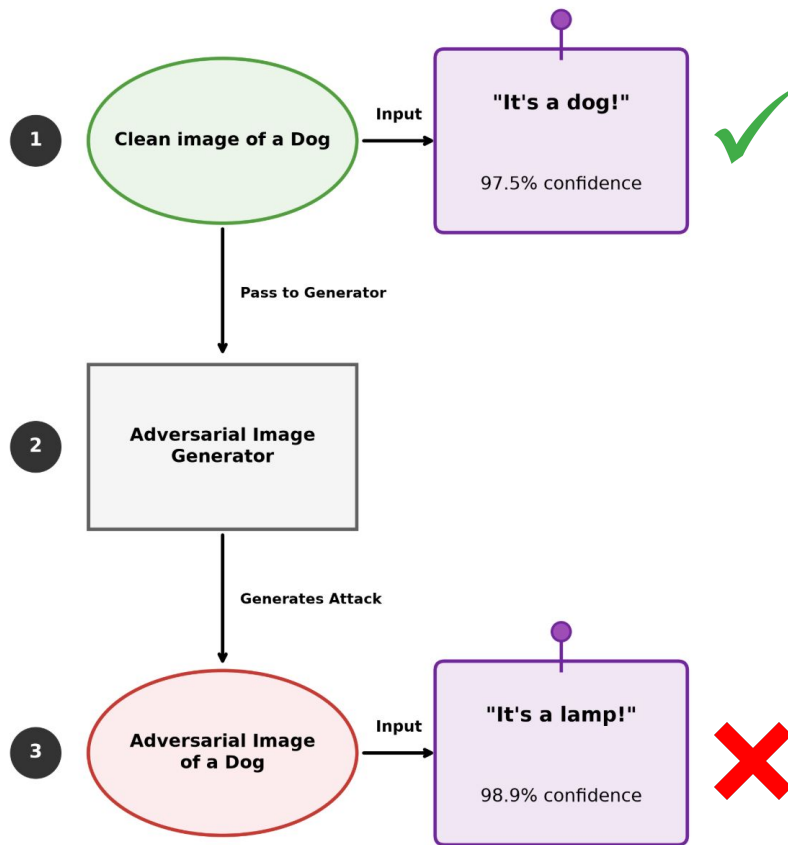
99.3 % confidence

Definition and Foundation

Adversarial Attack: An imperceptible input perturbation that tricks neural networks into high-confidence misclassifications (Goodfellow et al., 1).

Root Cause: Driven by model linearity in high-dimensional space rather than non-linear unpredictability (Goodfellow et al., 2).

How? Because being predictable also means being easily exploitable by aligned perturbations.



Generate Adversarial Images with FGSM

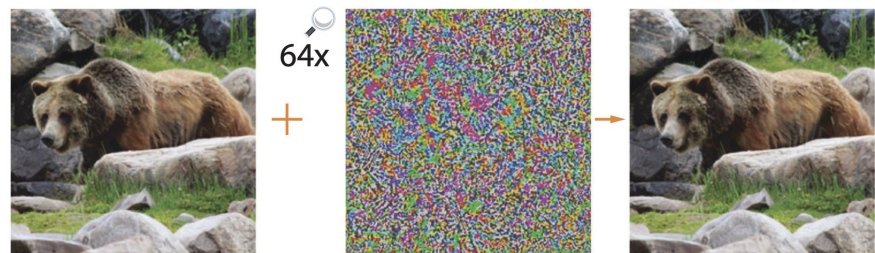
$$\eta = \epsilon \cdot \text{sign}(\nabla_x J(\theta, x, y)), \quad \tilde{x} = x + \eta$$

where ϵ : how large the change is allowed to be,
 η : the resulting change added to the image,
 $\tilde{x}$: the final adversarial image.

Other Adversarial Examples

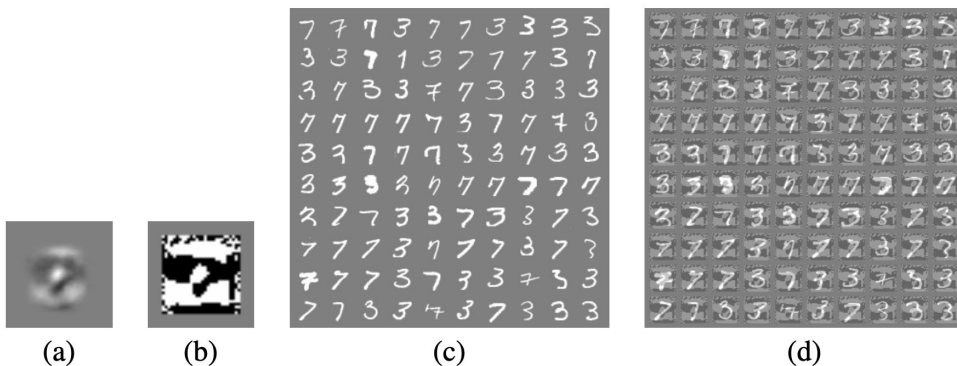
Ex. 2 Bear to Truck

illustration of a typical adversarial example
ANN classification: bear



(Veerabadran et al., 2)

Ex. 3 MNIST Logistic Regression



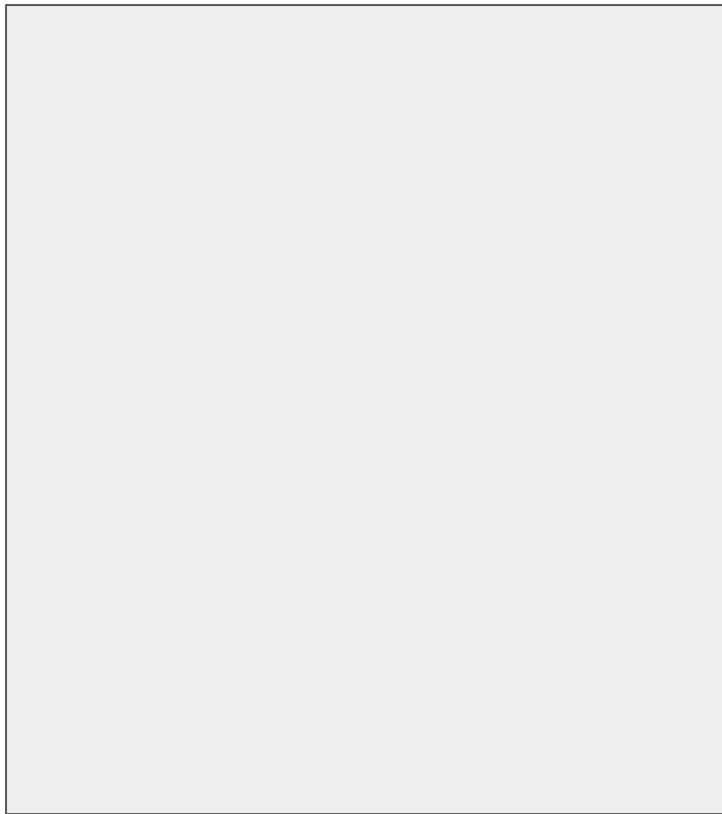
(Goodfellow et al., 4)

Why Does This Happen?

Quick Refresher: Issue is not nonlinear (unpredictable), but linearity (predictable) in high-dimensional spaces. (Goodfellow et al., 1)

- Every pixel input is being considered, a model's decision
- These adversarial changes point in the same direction
- They accumulate instead of canceling out
- FGSM finds direction that makes those changes stack up

Result: Small, calculated changes invisible to human eyes, but completely overthrows the model's output precision.

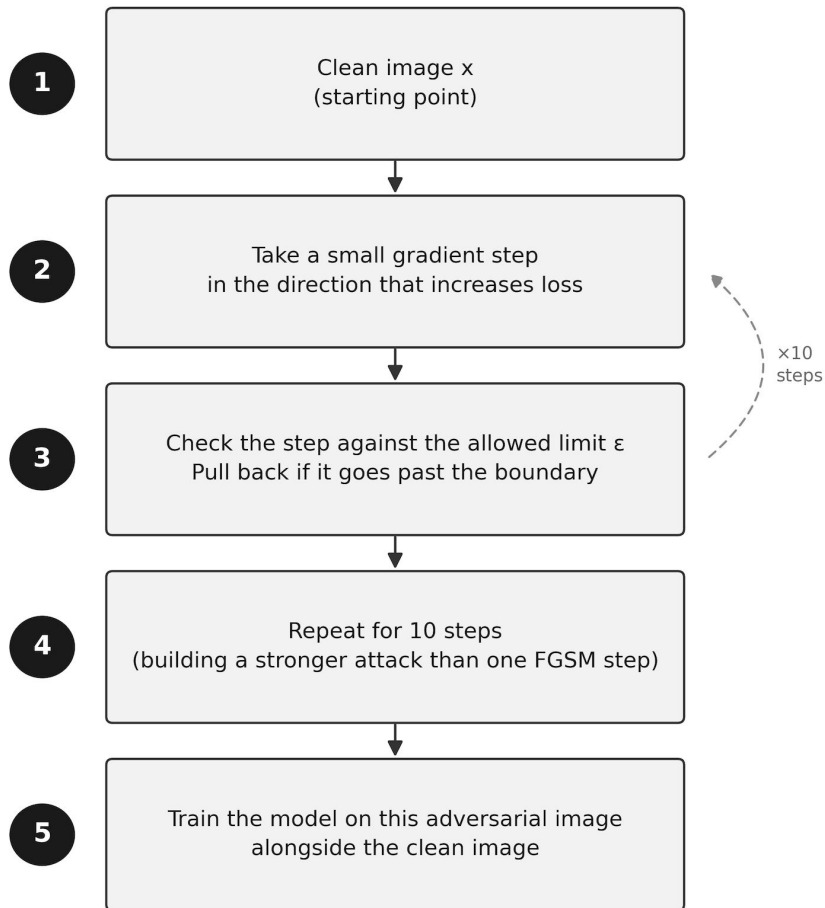


How Can We Defend From This?

Adversarial Training: Train on both clean and adversarial images.

PGD (Projected Gradient Descent) vs. FGSM (Fast Gradient Sign Method): FGSM takes one step. PGD takes several small steps, checking the boundary each time. Rice et al. use PGD rather than FGSM.

Step-by-step on how PGD adversarial training works is shown on right side.



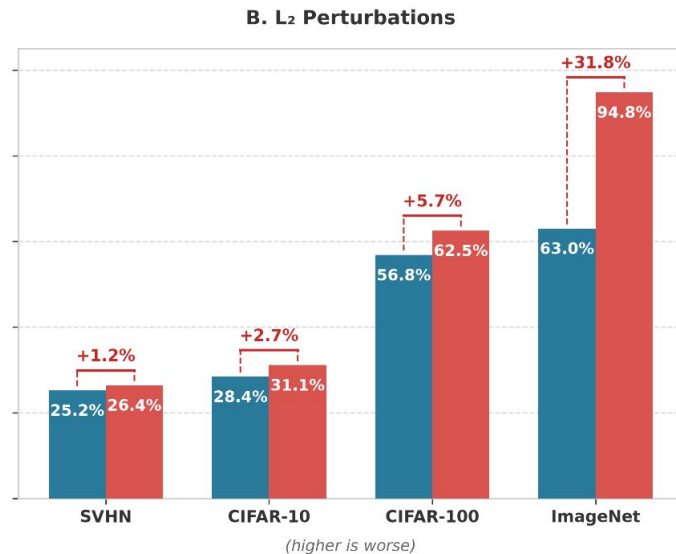
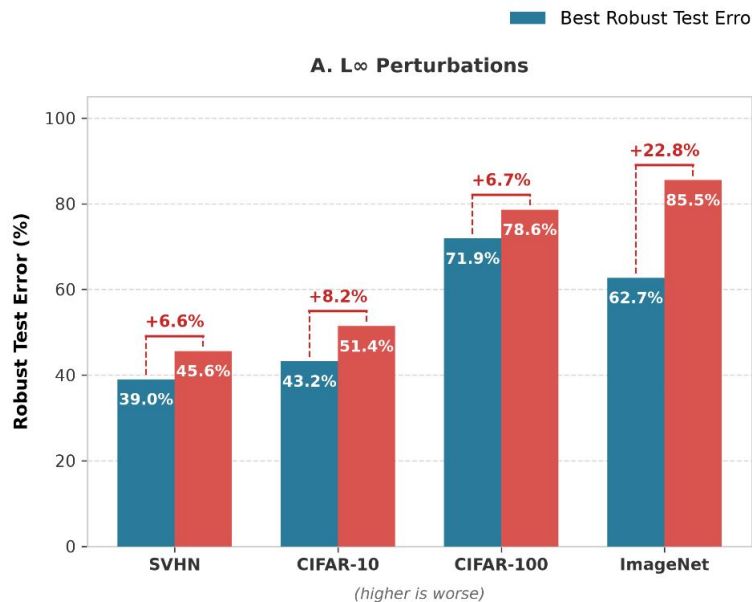
Training Too Long Can Hurt Robustness (Rice et al., 4-5)

Robust overfitting:

1. Robustness peaks early
2. Continued training erodes those gains

Robust test error: % of attacked test images misclassified.

Training longer does not mean more robust.

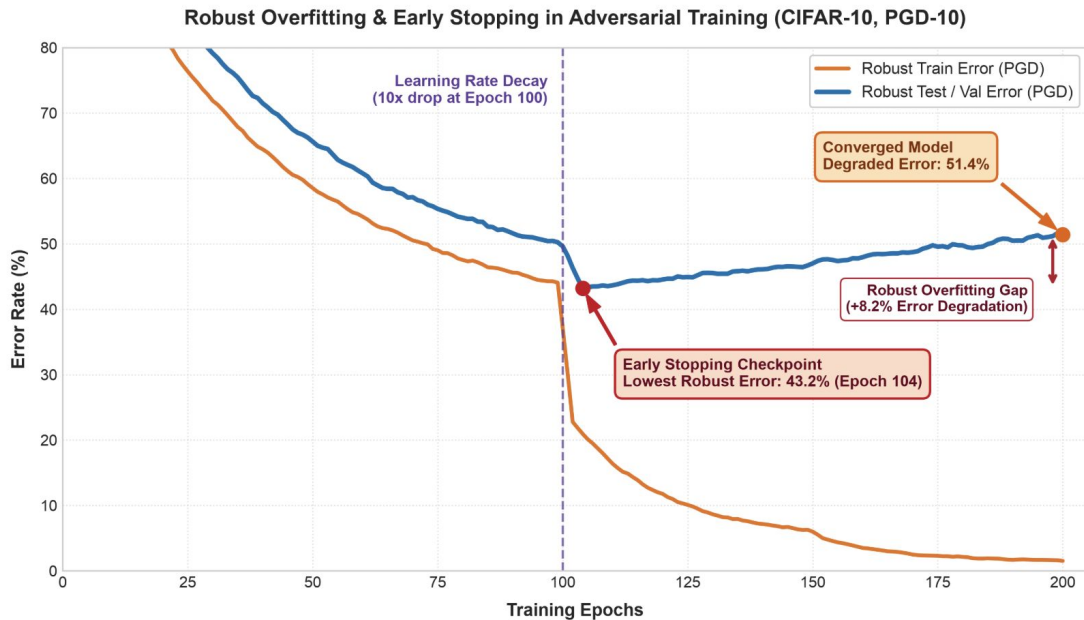


Solution to Robust Overfitting: Early Stopping

Core Mechanism: Save the best checkpoint evaluated on a held-out adversarial validation set rather than training until convergence.

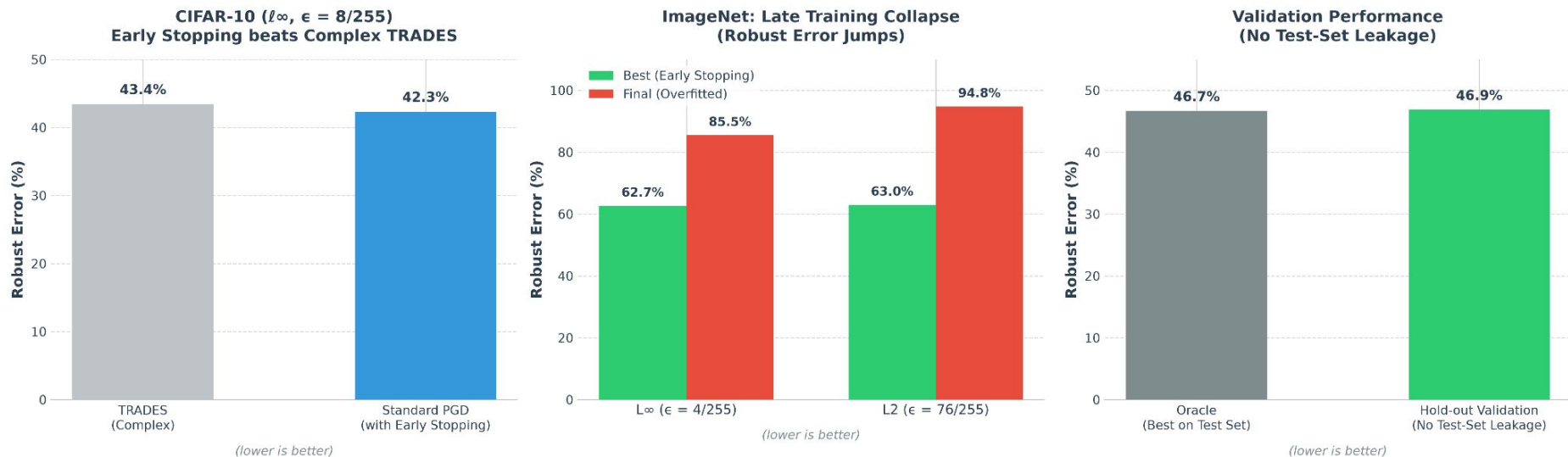
Preventing Degradation: Halt training shortly after initial learning rate decay to avoid severe robust test set overfitting.

Proven Effectiveness: Simple early stopping matches or outperforms complex explicit regularization techniques across architectures.



(Rice et al., 1, 6)

Effectiveness of Early Stopping (Rice et al., 4)



Our Research: Motivation and Scope

Open Question:

Why does robust overfitting happen internally? This was not explained in Rice et al. paper.

Why It Matters:

A cause-level answer could replace early stopping with a real safeguard.

Scope of This Research:

Reproduce Rice et al. findings on CIFAR-10, track robust accuracy by epoch.

Upcoming Slides:

We will show our research reproduction progress in the upcoming slides.

Adversarial Generation and Training Loop Algorithm

[Text goes here]

Finding on Training and Testing

[Text goes here]

Finding on Early Stopping

[Text goes here]

All code for reproducing the experiments as well as pretrained model weights and training logs can be found at GitHub repository: github.com/ItsKaiwenDu/Robust-Overfitting.

Key Takeaways

1. Adversarial examples, which can be generated with FGSM, are caused by linearity (predictable so exploitable), not nonlinearity (unpredictable), to reliably trick confident models.
2. Adversarial training defends against adversarial attacks by training on both clean and perturbed examples.
3. Robust overfitting causes adversarially trained models to get worse on new data the longer they train.
4. Early stopping is the best and simplest way to resist robust overfitting and this method matches or beats more complex methods.
5. [Reserved for our research findings]
6. [Reserved for our research findings]

References

Goodfellow, Ian J., Jonathon Shlens, and Christian Szegedy. "Explaining and Harnessing Adversarial Examples." *International Conference on Learning Representations (ICLR)*, 2015.

Rice, Leslie, Eric Wong, and J. Zico Kolter. "Overfitting in Adversarially Robust Deep Learning." Proceedings of the 37th International Conference on Machine Learning, *PMLR*, vol. 119, 2020.

Veerabadran, Vijay, et al. "Subtle Adversarial Image Manipulations Influence Both Human and Machine Perception." *Nature Communications*, vol. 14, no. 1, 2023, article 4933. Springer Nature, 15 Aug. 2023, <https://doi.org/10.1038/s41467-023-40499-0>

Du, Kaiwen, and Nicholas Q. Tran. Investigating Robust Overfitting in Adversarial Training. *GitHub*, 2026, github.com/ItsKaiwenDu/Robust-Overfitting.



Thank you!

Any Questions?

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